

**γ -hadron correlations with only CNM effect in $Pb + Pb$ collisions
at $\sqrt{s_{NN}} = 5.02$ TeV**

• **Setting for the numerical calculations**

TABLE I. Configurations for the NLO pQCD calculations.

system	$Pb + Pb$ collisions
colliding energy	$\sqrt{s_{NN}} = 5.02$ TeV
rapidity	$ y^\gamma < 0.67, y^h < 0.9$
away-side range	$ \Delta\phi > 3\pi/5$
$p_T^{\text{trig}}(\gamma)$	[18-40] [GeV/ c]
$p_T^{\text{assoc}}(h^\pm)$	[> 0.5] [GeV/ c]
z_T	[0.1-0.15], [0.15-0.2], [0.2-0.3], [0.3-0.4], [0.4-0.6], [0.6-0.8], [0.8-1.0]
PDFs	CT18A PDF [1]
nPDFs	EPPS21 nPDF [2]
FFs	Kniehl-Kramer-Potter FFs[3]
	Without jet energy loss

• **Cold nuclear effect with uncertainty**

The nuclear parton distribution functions (nPDFs) $f_{a/A}(x_a, \mu^2, \vec{r})$ can be expressed as [4–7],

$$f_{a/A}(x_a, \mu^2, \vec{r}) = S_{a/A}(x_a, \mu^2, \vec{r}) \left[\frac{Z}{A} f_{a/p}(x_a, \mu^2) \left(1 - \frac{Z}{A} \right) f_{a/n}(x_a, \mu^2) \right], \quad (1)$$

where Z is the charge number of the nucleus. The averaged nuclear modification factor $S_{a/A}(x_a, \mu^2)$ is taken from the EPPS21 parameterization.

The EPPS21 nPDFs use the Hessian-matrix based approach to estimate the PDF uncertainties. Let us consider one quantity X that depends on the PDFs and $X(S_k^\pm)$ denotes the value of X computed with the PDF set $S_k^\pm$. The upper and lower uncertainties ΔX^+ and ΔX^- can be obtained by the following prescription,

$$(\Delta X^+)^2 \approx \sum_k [\max\{X(S_k^+) - X(S^0), X(S_k^-) - X(S^0), 0\}]^2 \quad (2)$$

$$(\Delta X^-)^2 \approx \sum_k [\max\{X(S^0) - X(S_k^+), X(S^0) - X(S_k^-), 0\}]^2. \quad (3)$$

where S^0 denotes the central PDF set giving the minimum χ^2 .

In EPPS21 nPDFs, pset= 1...107,

1 = central fit

2,3 = error sets S_1^+, S_1^-

4,5 = error sets S_2^+, S_2^-

... ..

106,107 = error sets S_{53}^+, S_{53}^- .

• Numerical results

The cold nuclear matter effects reflected by γ -hadron yield and direct- γ production could largely cancel out. I_{AA}^h can also be expressed as [8, 9],

$$I_{AA}^h(z_T) = \frac{J_{AA}^h(z_T)}{\langle R_{AA}^\gamma(p_T) \rangle}, \quad (4)$$

where J_{AA}^h is the ratio of γ -hadron yield in $A + A$ collisions over that in $p + p$ collisions,

$$J_{AA}^h(z_T) = \frac{\frac{d\sigma_{AA}^{\gamma h}}{dy^\gamma dp_T^\gamma dy^h dp_T^h d\phi}}{\langle N_{\text{binary}}^{AA} \rangle \frac{d\sigma_{pp}^{\gamma h}}{dy^\gamma dp_T^\gamma dy^h dp_T^h d\phi}}. \quad (5)$$

The average $\langle R_{AA}^\gamma(p_T) \rangle$ over a p_T range from 18 to 40 GeV/c is calculated as,

$$\langle R_{AA}^\gamma(p_T) \rangle = \frac{\langle d\sigma_{AA}^\gamma / dy^\gamma d^2p_T^\gamma \rangle}{\langle N_{\text{binary}}^{AA} \rangle \langle d\sigma_{pp}^\gamma / dy^\gamma d^2p_T^\gamma \rangle}. \quad (6)$$

Figure 1 illustrates the uncertainty arising from the factorization scale. The first panel shows the γ -triggered fragmentation functions (D_{pp}) in $p+p$ collisions at 5.02 TeV for factorization scales $\mu = 0.7, 1.2, 2.0p_T^{\text{trig}}$. The second panel presents the γ -triggered fragmentation functions (D_{AA}) with only the central value of the cold nuclear matter (CNM) effect in $Pb + Pb$ collisions at 5.02 TeV, also for factorization scales $\mu = 0.7, 1.2, 2.0p_T^{\text{trig}}$. The corresponding nuclear modification ratio $I_{AA}(\text{CNM}) = D_{AA}/D_{pp}$ is shown in the third panel. It can be observed that the uncertainty from the scale μ is almost completely cancelled in I_{AA} . Based on our experience from the best fit to γ production, we set $\mu = 1.2p_T^{\text{trig}}$.

Figure 2 shows the uncertainty from cold nuclear matter effects, with the factorization scale fixed at $\mu = 1.2p_T^{\text{trig}}$. The left panel displays the γ -triggered fragmentation functions D_{AA} including only the CNM effect: the solid blue curve represents the central value, and the blue shaded band indicates the associated uncertainty. The right panel presents the

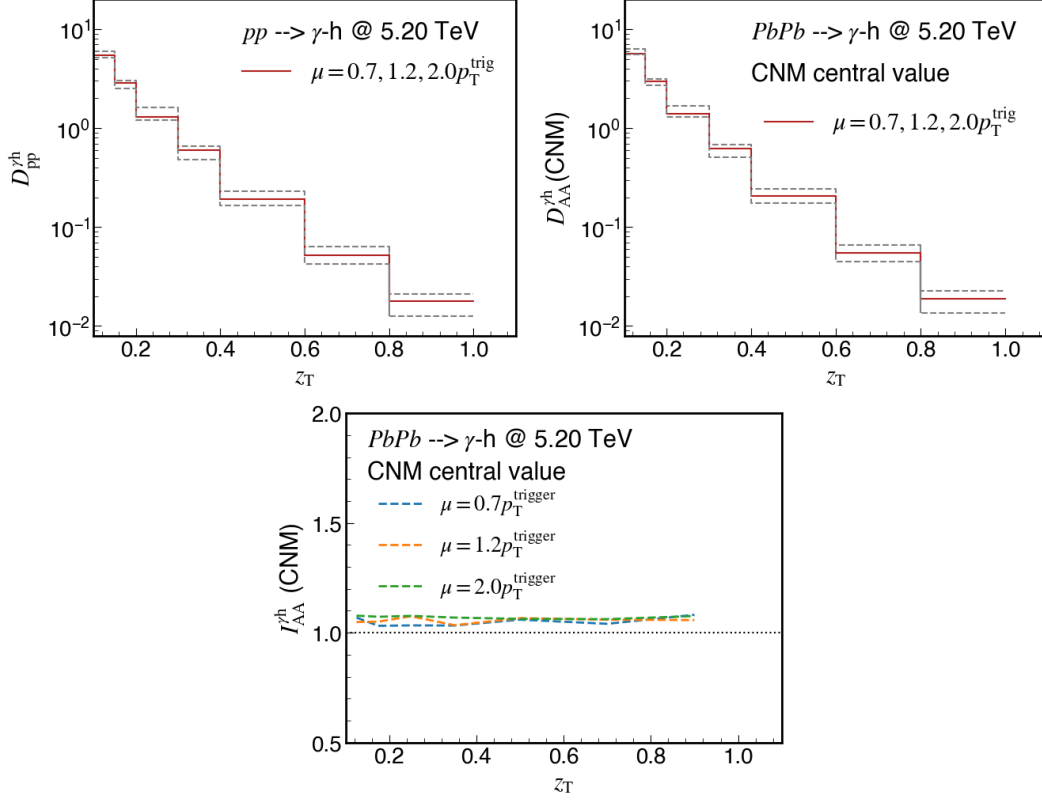


FIG. 1. γ -triggered fragmentation functions in $p + p$ collisions and $Pb + Pb$ collisions with only central value of CNM effect at $\sqrt{s_{NN}} = 5.02$ TeV. The red curve is the NLO pQCD result for factorization scale $\mu = 1.2 p_T^{\text{trig}}$. The gray dashed curves are the results for $\mu = 0.7$ and $2.0 p_T^{\text{trig}}$ to show the uncertainty from factorization scale. The corresponding $I_{AA}(\text{CNM}) = D_{AA}/D_{pp}$ is shown in the third plot.

nuclear modification factor R_{AA} for photons with $18 < p_T^\gamma < 40$ GeV/ c (blue), the γ -hadron correlation $J_{AA}(\text{CNM})$ (green), and the modification factor of γ -hadron correlation $I_{AA}(\text{CNM})$ (red). In our experience, the uncertainties in $I_{AA}(\text{CNM}) = J_{AA}/R_{AA}$ should largely cancel. However, the NLO pQCD calculation for γ -hadron correlations still exhibits relatively large uncertainties. For a fixed nPDF set, I typically perform multiple calculations by varying the number of sample points in the Vegas integration and the random-number sequences, then average the results to improve the accuracy of the theoretical prediction. This approach, however, is too difficult to implement for 107 nPDF sets. Furthermore, we have not yet systematically evaluated or addressed the uncertainties introduced by the pQCD calculation itself.

This is also what I previously mentioned: even if we provide the uncertainties for the CNM effect in D_{AA} and I_{AA} , these may not represent the true uncertainty of the CNM effect itself, but rather largely reflect the uncertainty arising from theoretical computational errors. Moreover, we are currently unable to eliminate the uncertainty introduced by NLO pQCD calculations for the γ -hadron yield.

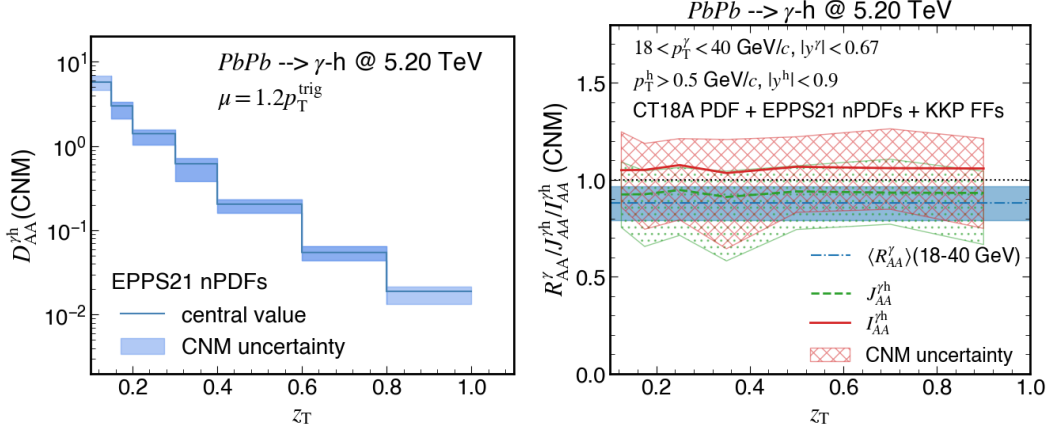


FIG. 2. Left: γ -triggered fragmentation functions with only CNM effect for $Pb + Pb$ collisions at $\sqrt{s_{NN}} = 5.02 \text{ TeV}$. Right: $R_{AA}/J_{AA}/I_{AA}$ with only CNM effect for $Pb + Pb$ collisions at $\sqrt{s_{NN}} = 5.02 \text{ TeV}$. The solid or dashed curves are the results for central values of the cold nuclear matter effect. The shaded bands are the results for uncertainty from 107 sets EPPS21 nPDFs.

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